

12(a)	- frequency determines energy of photon - intensity determines number of photons (per unit time) - intensity does not determine energy of a photon <i>Any two points, 1 mark each</i>	B2
	kinetic energy (of the electron) depends on the energy of one photon	B1
12(b)(i)	$E = hc / \lambda$ or $E = hf$ and $c = f\lambda$	C1
	$E = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (250 \times 10^{-9})$	C1
	$(= 7.96 \times 10^{-19} \text{ J})$ $= 5.0 \text{ eV}$	A1
12(b)(ii)	$E_{\text{MAX}} = \text{photon energy} - \text{work function}$	C1
	work function = $5.0 - 1.4$ $= 3.6 \text{ eV}$	A1